

PRS XVA Model (MKM-XV-001)

Credit, Funding and Capital Valuation Adjustments on the Physical Risk Swap

MKM Research Labs

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1 Executive Summary

MKM-XV-001 adds credit, funding and capital valuation adjustments (CVA, FVA, KVA, with DVA disclosed) to the Physical Risk Swap. It follows Green, *XVA: Credit, Funding and Capital Valuation Adjustments* (Wiley, 2015): every adjustment is a spread times a survival-discounted expected exposure, and the funding strategy decides which terms exist. The model is an overlay on the PRS pricer (MKM-PR-001): it reproduces the pricer's value at $t = 0$ and does not change it.

Status. ILLUSTRATIVE, version 0.10, submitted for independent validation. Counterparty credit, the bank's funding spread and CSA terms are placeholders; the regulatory treatment that drives KVA is undecided. KVA is reported beside the booked XVA and never inside it (decision D-16).

Findings. On a GBP 10m, 5-year PRS with a 2% annual trigger probability:

1. Under independence, CVA and FVA are below 1bp of running spread. The protection buyer's CVA is almost entirely the unpaid notional during the settlement lag.
2. Hazard volatility and wrong-way risk move CVA by a factor of 4–13, so their calibration matters more than any credit input.
3. Capital dominates: KVA is about two orders of magnitude larger than CVA and FVA (Table 7), about 43bp of running spread against a fair spread of 203.56bp. The regulatory treatment of a PRS is worth about 32bp of spread on this trade, and the RRAO a further 15bp.
4. Variation margin does not reach the jump to notional; initial margin does. A CSA is worth far more in capital than in CVA.

Evidence. All six benchmark checks pass (Table 3), including agreement with an independent re-implementation to better than 10^{-8} and the Basel Committee's own SA-CCR worked examples. Figures are from `docs/xva/validation_evidence.py` at commit `fdcd37c0672e` (10,000 paths, config `301a1ed41a80`, generated 2026-10-12T12:55:02+00:00): all pass.

2 Model Purpose and Scope

2.1 What the model answers

What a PRS is worth to the bank once the counterparty's default, the bank's own funding and the regulatory capital the trade consumes are priced in; and therefore the all-in spread at which the bank should deal.

2.2 The instrument

A PRS is a single-name CDS with a flood trigger in place of the credit event. The protection buyer pays a quarterly running spread in arrears while no trigger has occurred; the seller pays the full notional (recovery 0) when the reference gauge first passes its Severe level. The XVA model takes its contract terms from the pricer (D-10), including first-trigger termination, and adds a settlement lag Δ_s (provisionally 5 business days, D-09) in which the payout is owed but unpaid.

2.3 Scope

- In scope: gauge PRS (`TradeType = 'PRS'`), bank as buyer or seller, netting sets of such trades, VM and IM under a CSA, a wrong-way-risk stress, SA-CCR and BA-CVA capital, RRAO as a switch.
- Out of scope: Property PRS (no term pricer, §8), MVA, TVA, warehousing (Green Ch. 14) and portfolio KVA (Ch. 15).

2.4 Model chain

Upstream: MKM-PR-001 (PRS value), MKM-EF-001 (event catalogue and arrival rate). Downstream: MKM-TD-001 (Trading Desk blotter and EOD). Shares its counterparty-capital module with MKM-CR-001 (D-07).

3 Mathematical Framework

3.1 Green's semi-replication

Write the all-in value as $\hat{V} = V + U$. Hedging the trade and both defaults with a self-financing portfolio and applying Feynman–Kac gives

$$U(t) = -\int_t^T D(t,s)\,\mathbb{E}_t[f(s)]\,ds, \tag{1}$$

with one source term per adjustment:

$$\text{CVA} = -(1-R_C)\int \lambda_C(s)\,D(0,s)\,\mathbb{E}[(V-X-\tilde{I})^+]\,ds \quad (3.70, 10.31) \tag{2}$$

$$\text{DVA} = -(1-R_B)\int \lambda_B(s)\,D(0,s)\,\mathbb{E}[(V-X+I)^-]\,ds \quad (3.71) \tag{3}$$

$$\text{FCA} = -(1-R_B)\int \lambda_B(s)\,D(0,s)\,\mathbb{E}[(V-X+I)^+]\,ds \quad (9.57, 10.28) \tag{4}$$

$$\text{FVA} = -\int s_F\,D(0,s)\,\mathbb{E}[V-X]\,ds \quad (9.64\text{–}9.67) \tag{5}$$

$$\text{KVA} = -\int (\gamma_K - \phi\,r(s))\,D(0,s)\,\mathbb{E}[K(s)]\,ds \quad (13.25) \tag{6}$$

3.2 Funding strategy and discounting (D-01, D-03)

The booked strategy is Green's one-bond strategy: unilateral CVA plus a symmetric FVA, with $D(0,s) = DF(s)\,e^{-s_F s}S_C(s)$ (discounting at $r_F + \lambda_C$). The semi-replication view, CVA + DVA + FCA with $D(0,s) = DF(s)\,S_B(s)\,S_C(s)$, is computed beside it and its DVA is disclosed. The two coincide exactly when $R_B = 0$, which is a test. D-03 as first written applied $r + \lambda_B + \lambda_C$ to every term; that is correct only for semi-replication and was corrected in WP1.

3.3 Why a PRS is different

Exposure is a jump, not a diffusion: before a trigger the value is a small spread MTM; after it, $\pm N$ until settlement. Under deterministic hazard the pre-trigger value is deterministic, so

$$\mathbb{E}[V(s)^+] = S_E(s)\,V(s)^+ + \int_{\max(0,s-\Delta_s)}^{\min(s,T)} \lambda_E S_E(u)\,P(u,s)^+\,du, \tag{7}$$

with $P(u,s)$ the pending payout. This is exact, and is the closed form the Monte Carlo must reproduce.

3.4 Stochastic flood hazard: CIR++ (D-15)

A catchment factor Y follows CIR with $Y(0) = \theta = 1$: $dY = \kappa(1-Y)\,dt + \sigma\sqrt{Y}\,dW$. Gauge g's trigger intensity is $\lambda_g Y(t) + \varphi_g(t)$, with the shift fitted so the model's survival at $t = 0$ is exactly the market curve (Brigo–Mercurio). The curve may be a term structure (§3.5); λ_g is its lowest hazard, which keeps the shift a non-negative intensity. Given $Y(t) = y$ and no trigger,

$$S(t,u) = e^{-(\Phi_g(u)-\Phi_g(t))}\,A(u-t)\,e^{-B(u-t)\lambda_g y}, \tag{8}$$

so $V(t)$ is closed form on every path. A is evaluated in a form that is stable as $\sigma \to 0$ (the textbook form loses precision there, which the tests caught). Y is stepped exactly by its noncentral χ^2 transition.

3.5 Hazard term structure (WP6)

On the desk each gauge's curve is bootstrapped, piecewise constant, from the market state's tenor-bucket hazards so that every bucket reprices at the front office's fair spread (recovery 0, pricer convention). A curve that cannot be bootstrapped (a negative forward hazard) falls back to flat at the longest bucket and is recorded in the run's audit block. On a 20-trade test book this took the median gap between engine and front-office value from 29bp to 0.05bp of notional.

3.6 Correlated triggers

The MKM-EF-001 catalogue supplies dependence: catchment events arrive as a Cox process in the factor clock $C(t) = \int_0^t Y$, are drawn by catalogue weight, and their Severe flags say which gauges they trigger. Each gauge's marginal is held to the market rate by thinning (catalogue rate above market) or a Y-driven top-up (below).

3.7 Collateral, netting and wrong-way risk

Variation margin follows Green's lookback (6.39) with threshold and minimum transfer amount over the margin period of risk; initial margin is static in each direction. Without close-out netting, exposures add trade by trade. The wrong-way-risk stress weights each path by $e^{\beta Z}/\mathbb{E}[e^{\beta Z}]$, Z the number of the netting set's gauges triggered so far, normalised so each period's default probability is unchanged (Green 7.17).

3.8 Capital (D-06, D-07, D-16)

SA-CCR EAD and BA-CVA (reduced) are computed in every Monte Carlo state (Green §13.4). How a PRS enters SA-CCR is a switch: commodity "other" (SF 18%) or a single-name credit analogy. Capital held is RWA times the bank's 10.5% ratio, shared with MKM-CR-001.

4 Input Parameters and Calibration

Parameter	Value	Source / status
Flood trigger hazard	market state	Per-gauge term structure from the desk's tenor buckets (§3.5)
Risk-free curve	market state	Desk yield curve (D-13)
Counterparty hazard	rating proxy	`RATING_CDS_SPREADS`: **placeholder** (D-11)
R_C, R_B	40%, 40%	Market convention
s_F	75bp	**Placeholder**, Treasury input (D-12)
κ, σ	0.5, 0.3	**Placeholder**, no calibration owner (D-15)
Δ_s	5 business days	**Provisional**, Product/Legal (D-09)
β (WWR)	0	Stress only (D-05)
γ_K, ϕ	12%, 0	Shared with MKM-CR-001 (D-06)
Capital ratio	10.5%	Shared with MKM-CR-001
SA-CCR treatment	commodity other	Regulatory paper Q2 (D-16)
RRAO	off	Regulatory paper Q3
Paths, seed	10,000, 20261009	D-08
Time cells	26 per quarter	D-14

5 Implementation Details

Module	Role
`config/xva.py`	Every parameter; frozen, fingerprinted
`models/xva/curves.py`	Pricer-identical discounting; hazard curves; CDS bootstrap
`models/xva/valuation.py`	$V(t)$, fair spread, pending payout
`models/xva/exposure_analytic.py`	Exact EPE/ENE/EE
`models/xva/integrals.py`	Kernels and integrals; the only place discounting lives
`models/xva/analytic.py`	WP1 engine; exact break-even spread
`models/xva/{cir,events,paths}.py`	CIR++, correlated triggers, grid and path values
`models/xva/netting.py`	CSA and netting sets from the CDM
`models/xva/monte_carlo.py`	WP2 engine; per-path estimates with SE; Euler allocation
`models/capital/counterparty.py`	SA-CCR, BA-CVA, risk weights (shared, D-07)

`models/xva/capital.py`	Capital per Monte Carlo state
`models/xva/desk.py`, `routes/trading/xva.py`	Trading Desk integration (WP4)

A 16-trade netting set runs in about 10 seconds at 10,000 paths; memory is $O(\text{paths} \times \text{nodes})$ whatever the number of trades. The desk runs XVA on demand, at EOD and for quotes; the blotter reads the stored run (D-17).

6 Validation and Backtesting

6.1 Benchmarks

Table 3: Benchmark checks

Check	Result	Tolerance	Outcome
Pricer reconciliation (fair spread, annuity)	4.2×10^{-16}	$< 10^{-12}$	pass
Independent re-implementation, all terms	3.7×10^{-11}	$< 10^{-8}$	pass
Monte Carlo against analytic, max $\lvert z \rvert$	1.56	< 3	pass
Time rule error at 26 cells / quarter	4.6×10^{-4}	$< 5 \times 10^{-4}$	pass
CIR++ fit to the market curve	2.2×10^{-16}	$< 10^{-12}$	pass
BCBS 279 Example 2 / 3 EAD	381 / 5406	381 / 5,406	pass

The independent re-implementation (`tests/models/xva/_reference.py`) shares no code with the engine and integrates adaptively. Green's two derivations of CVA, by expectation and by replication, agree within four standard errors when default times are sampled directly.

6.2 Monte Carlo against the closed form

Table 4: Monte Carlo ($\sigma = 0$) against the analytic engine

Bank	View	Term	MC	SE	Analytic	z
buys	one-bond	CVA	−1,974	4.7	−1,967	−1.31
buys	one-bond	FVA	−1,154	3.2	−1,150	−1.34
buys	one-bond	TOTAL	−3,128	7.9	−3,117	−1.32
buys	semi-replication	CVA	−1,960	4.7	−1,953	−1.31
buys	semi-replication	DVA	78	0.2	78	−1.56
buys	semi-replication	FCA	−1,225	2.9	−1,221	−1.31
buys	semi-replication	TOTAL	−3,107	7.8	−3,096	−1.32
sells	one-bond	CVA	−127	0.4	−128	+1.56
sells	one-bond	FVA	1,154	3.2	1,150	+1.34
sells	one-bond	TOTAL	1,027	3.6	1,022	+1.37
sells	semi-replication	CVA	−124	0.4	−125	+1.56
sells	semi-replication	DVA	1,225	2.9	1,221	+1.31
sells	semi-replication	FCA	−78	0.2	−78	+1.56
sells	semi-replication	TOTAL	1,023	3.5	1,018	+1.37

6.3 Results

Table 5: Analytic engine: GBP 10m, 5y, 2% trigger, at the fair spread

Ctpy	Bank	CVA	FVA	Booked	DVA	Semi-rep	bp
A	buys	−91	663	573	778	566	+0.14
A	sells	−574	−663	−1,237	123	−1,223	+0.30
BBB	buys	−192	647	454	759	449	+0.11
BBB	sells	−1,219	−647	−1,866	120	−1,844	+0.46
BB	buys	−448	603	156	708	154	+0.04
BB	sells	−2,834	−603	−3,438	112	−3,399	+0.88

Table 6: Monte Carlo engine, BBB-proxy counterparty

Case	CVA ± SE	FVA	Booked
buys, sigma 0	−201 ± 6	639	437
buys, sigma 0.3	−933 ± 11	653	−279
buys, WWR beta 3	−2,677 ± 75	653	−2,024
buys, VM	−622 ± 5	13	−610
buys, VM + IM = notional	−1 ± 0	13	12
sells, sigma 0.3	−1,970 ± 13	−653	−2,623
sells, WWR beta 3	−1,183 ± 8	−653	−1,837
sells, VM	−643 ± 5	−13	−656

Table 7: Capital and KVA, bank buys protection; bp is $|\text{KVA}|/(NA)$ of running spread

Treatment	EAD today	Capital today	KVA ± SE	bp
commodity other	2,520,000	538,129	−183,869 ± 267	43.08
credit analogy (ref. B)	990,972	211,616	−46,463 ± 61	10.89
commodity + RRAO	2,520,000	669,379	−247,527 ± 373	58.00
commodity, VM	756,000	161,439	−57,677 ± 88	13.52

6.4 Automated tests

Suite	**Purpose**
`tests/models/xva`	Curves, valuation, benchmark, limiting cases, CIR, events, netting, Monte Carlo, capital, desk
`tests/models/capital/test_counterparty.py`	BCBS 279 Annex 4a examples
`tests/routes/trading/test_xva_routes.py`	Run, blotter, quote, EOD
`tests/js/blotter-table.test.js`	XVA and KVA columns
`tests/port/cdm/test_credit_support.py`	Netting and CSA CDM fields

Coverage of `models.xva` and `models.capital` is 100%.

6.5 Outcomes analysis

Triggers are rare, so exposures cannot be backtested meaningfully. Outcomes analysis rests on the MKM-EF-001 hazard backtests, the daily valuation KPI (engine $V(0)$ against the front-office MTM, 5bp of notional) and the ΔXVA P&L line at EOD.

7 Sensitivity Analysis

Table 9: Sensitivities: booked XVA and KVA (GBP)

Case	Buys XVA	Buys KVA	Sells XVA	Sells KVA
base	−279	−183,869	−2,623	−187,668
counterparty hazard x0.5	193	−187,359	−1,676	−191,236
counterparty hazard x2	−1,166	−177,169	−4,394	−180,820
R_C 20%	−590	−183,869	−3,280	−187,668
R_C 60%	32	−183,869	−1,967	−187,668
s_F 0bp	−947	−186,498	−2,003	−190,355
s_F 150bp	366	−181,294	−3,221	−185,036
sigma 0	437	−182,235	−1,855	−186,002
sigma 0.6	−1,324	−184,497	−3,762	−188,455
kappa 0.25	−887	−184,091	−3,267	−187,941
kappa 1.0	221	−183,108	−2,121	−186,910
WWR beta 1	−523	−183,869	−2,495	−187,668
settlement 0d	31	−183,185	−2,738	−187,193
settlement 20d	−1,239	−183,958	−2,248	−187,103
credit-analogy SA-CCR	−279	−46,463	−2,623	−50,162
RRAO on	−279	−247,527	−2,623	−251,326
gamma_K 8%	−279	−122,579	−2,623	−125,112
gamma_K 15%	−279	−229,836	−2,623	−234,585
phi = 1	−279	−120,294	−2,623	−122,805

The base case is the trade above at the fair spread against a BBB-proxy counterparty, $\sigma = 0.3$, commodity SA-CCR. KVA is the larger figure in every row; among XVA inputs, hazard volatility and the settlement lag move the buyer most, counterparty hazard the seller.

8 Model Limitations and Known Weaknesses

L-1	Classical MPoR freezes collateral but not trade cash flows: a buyer that posted VM is briefly over-collateralised when it pays premium (Andersen–Pykhtin–Sokol).
L-2	The event clock integrates Y by the trapezoid rule; exact at $\sigma = 0$.
L-3	WWR driver is the netting set's triggered gauges, not the counterparty's real flood book.
L-4	Euler allocation is exact only for uncollateralised netted sets; residuals reported.
L-5	*Resolved in WP6:* CIR++ fits a hazard term structure. A steeper curve leaves a smaller share of hazard in the shared, stochastic part.
L-6	One seed, no variance reduction; the closed form is available as a control variate.
L-7	Every regulatory treatment is provisional; KVA is not booked.
L-8	The credit analogy needs a "reference rating" for a flood trigger.
L-9	BA-CVA applies CRE32's one-year maturity floor; to be confirmed.
L-10	Standalone BA-CVA: no diversification across counterparties.
L-11	No market-risk capital beyond the RRAO switch.
L-12	γ_K is a single constant.

L-13	*Resolved in WP6:* the desk bootstraps each gauge's term structure.
L-14	KVA allocated to trades pro rata to notional.
L-15	Netting set = counterparty (first CDM netting set).
L-16	No quote panel or "Run XVA" button in the UI yet.
L-17	Exercised on a generated 20-trade book on the local test portfolio; not yet on the live book.

Findings against other models. F-1: the pricer's MTM uses a fair spread floored at 2bp. F-2: the pricer ignores the settlement lag. F-3: the pricer rounds MTM to the penny. The trade record carries a 6M premium frequency while the pricer prices quarterly (D-10). F-4: the Trading Desk rounds remaining tenor to whole years (`_remaining_tenor`), dropping or adding up to a quarter of premium and protection; it is the only remaining source of valuation-KPI breaches on the test book.

9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

9.1 Q1 — Ronseal Test

"Does the model do what it says on the tin?"

Yes for the scope above. It reproduces the pricer at $t = 0$ and Green's integrals to 10^{-8}; its Monte Carlo reproduces the closed form.

9.2 Q2 — Accuracy and Stability

"Does the model provide timely, accurate and stable results?"

Accurate against every benchmark. Every Monte Carlo figure carries its standard error; results are reproducible from the seed. Runs take seconds per netting set.

9.3 Q3 — Resource Usage

"Does the model consume excessive computational resources?"

No: about 1s per trade at 10,000 paths, memory independent of trade count. It is kept out of per-request revaluation for this reason.

9.4 Q4 — Override Controls

"Are there controls over manual overrides?"

No manual overrides exist. Every parameter is in `config/xva.py`, fingerprinted into each run's audit block.

9.5 Q5 — Output Consumption

"How is the output consumed for risk decision-making?"

Blotter XVA and KVA memo columns, EOD ΔXVA, and incremental quotes. Nothing reaches a price or P&L until validation, and KVA not until the regulatory paper is answered.

9.6 Q6 — Model Chaining

"Is the model's output used as input to another model?"

Yes: it feeds MKM-TD-001, and is fed by MKM-PR-001 and MKM-EF-001; it shares capital code with MKM-CR-001.

9.7 Q7 — Weaknesses

"Does the model have weaknesses and limitations of use?"

Yes: §8. The material ones are placeholder inputs (credit, funding, σ), the open regulatory treatment, and L-1.

9.8 Q8 — Monitoring

"What type of monitoring is required?"

Daily valuation KPI; standard error per component; ΔXVA P&L explain; carry artefacts (grid roll, seed); annual recalibration of σ, κ.

9.9 Q9 — Overall Risk

"Is the overall model risk acceptable?"

Not yet for use: Tier 1, Amber. Acceptable for validation, with inputs and the regulatory treatment as gating items.

10 Document History

Date	Version	Description	Author
9-Oct-2026	0.9	Built (WP0–WP4); submitted for validation	David K Kelly
12-Oct-2026	0.10	Hazard term structure (WP6); F-4	David K Kelly